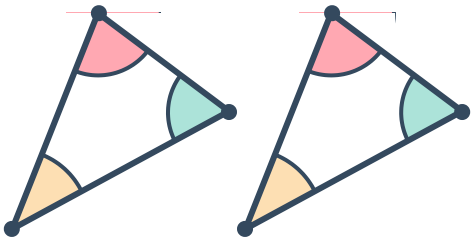
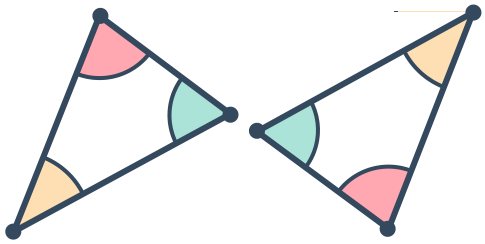
Rigid transformations

1 State whether the following pairs of triangles are a translation, rotation or reflection of one another:

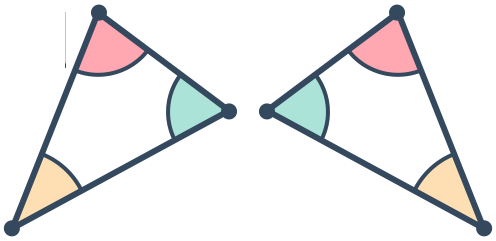
a



b

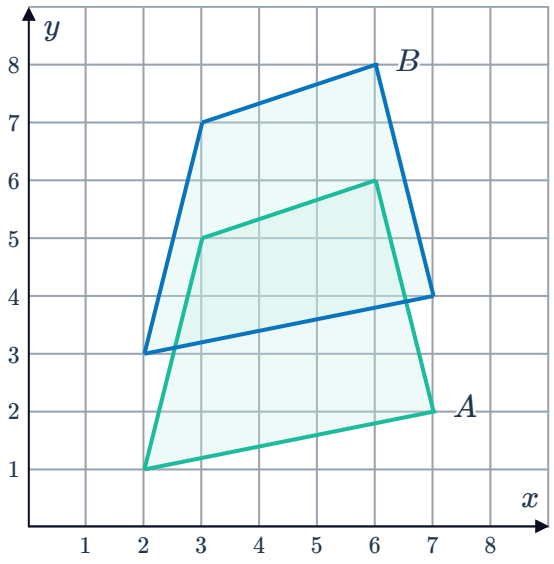


c

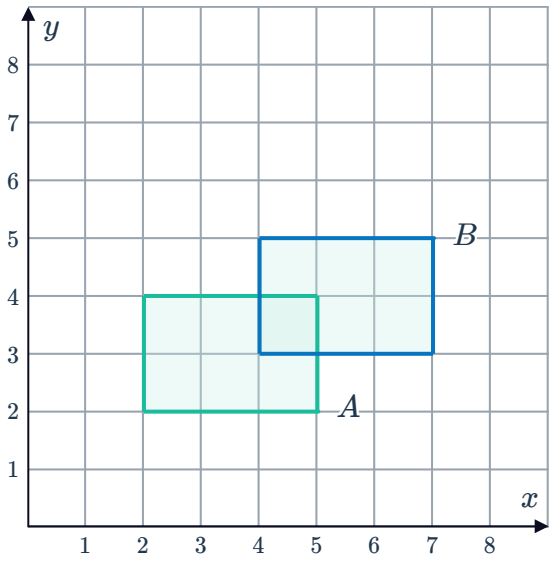


2 For each of the following, describe the translation from figure A to figure B:

a

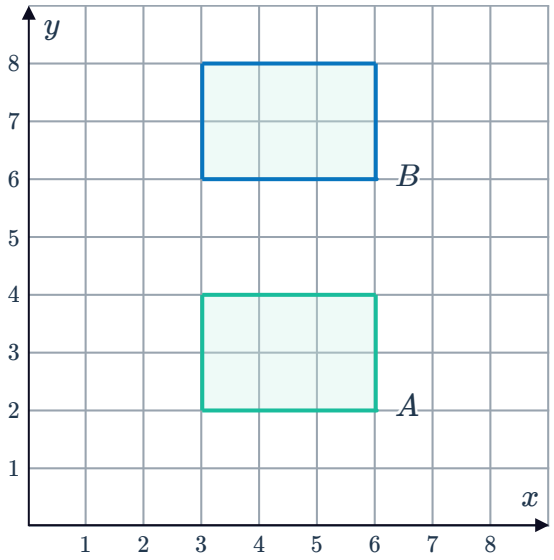


b

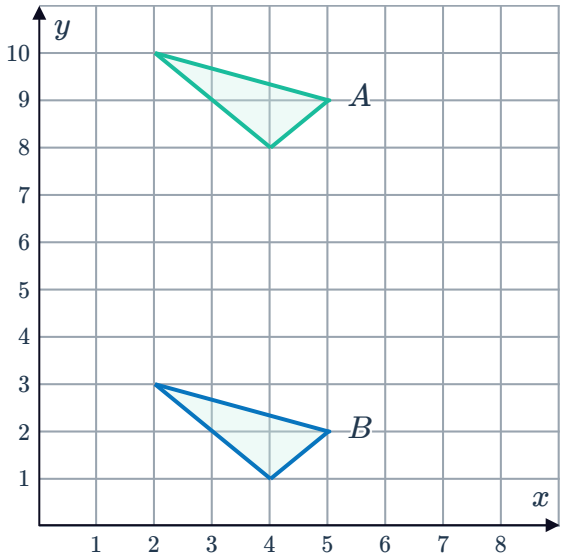


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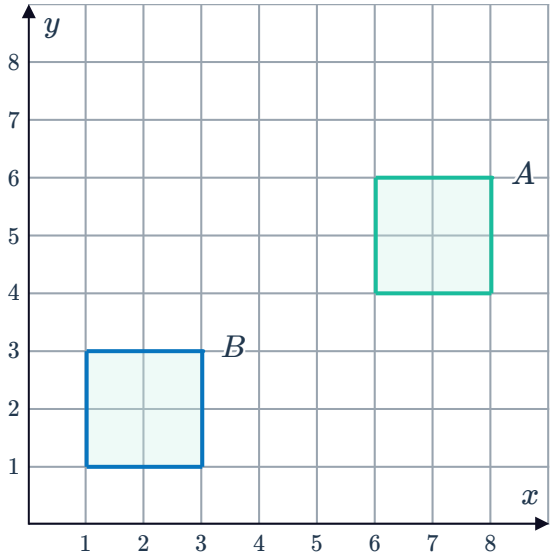
c



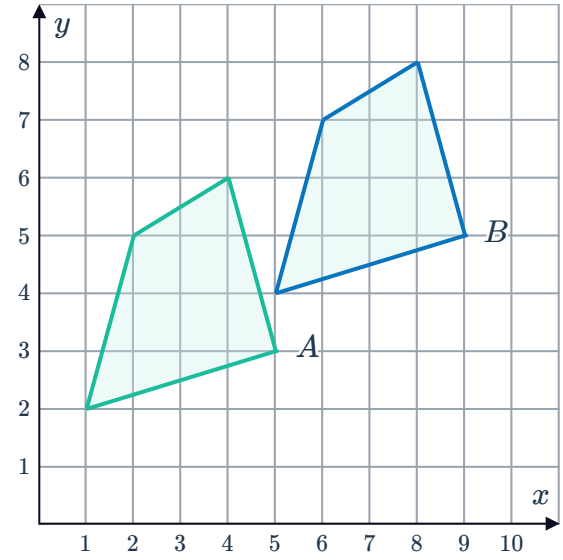
d



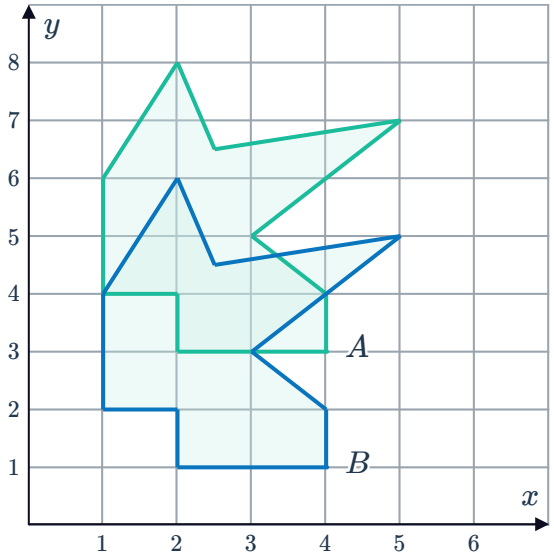
e



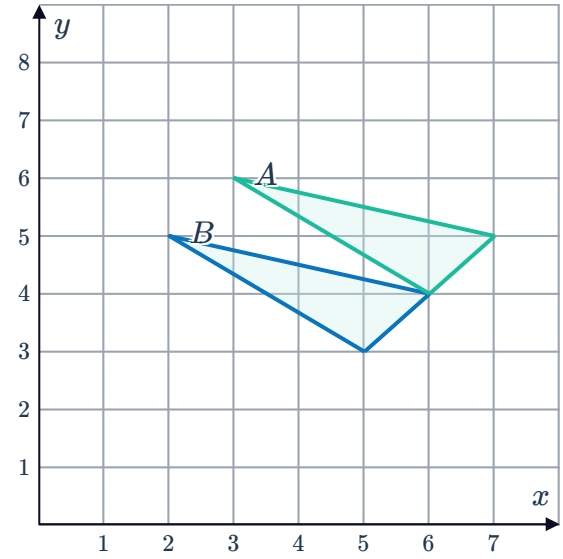
f



g



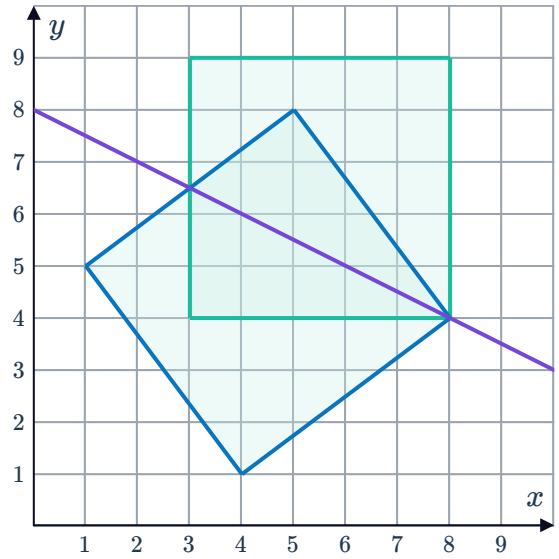
h



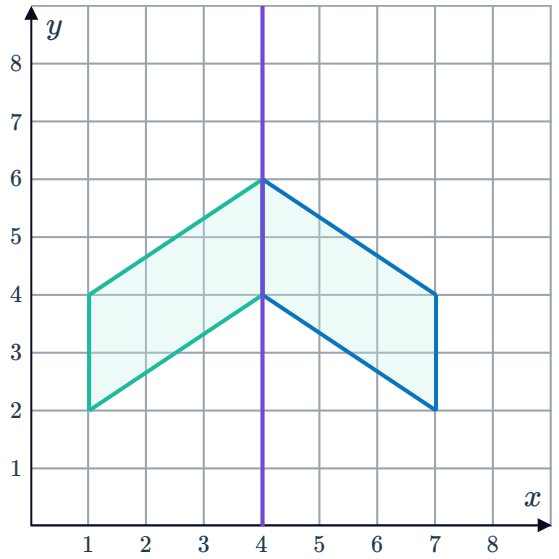
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3 Determine whether the following diagrams show a reflection across the given line:

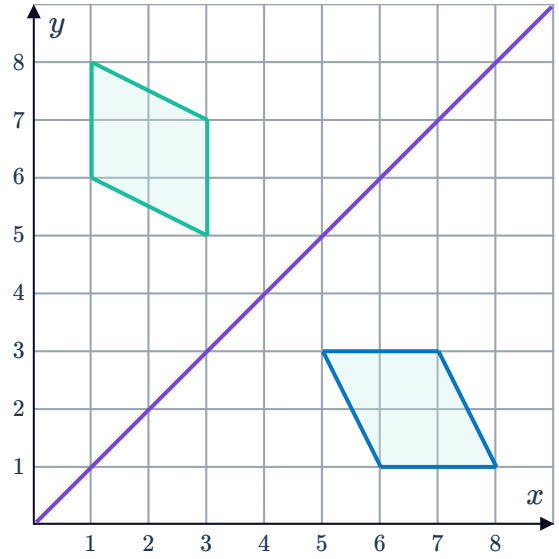
a



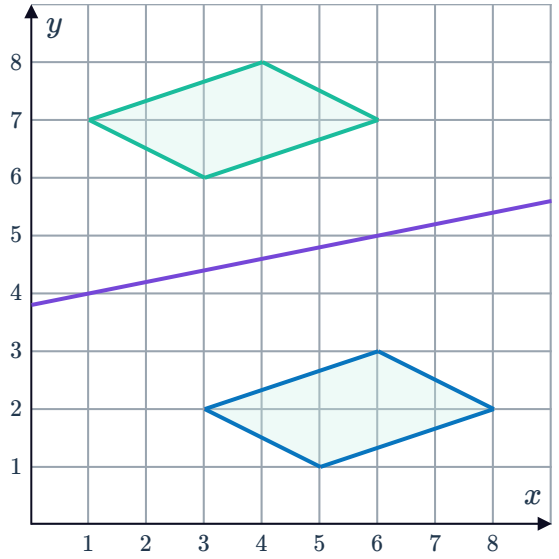
b



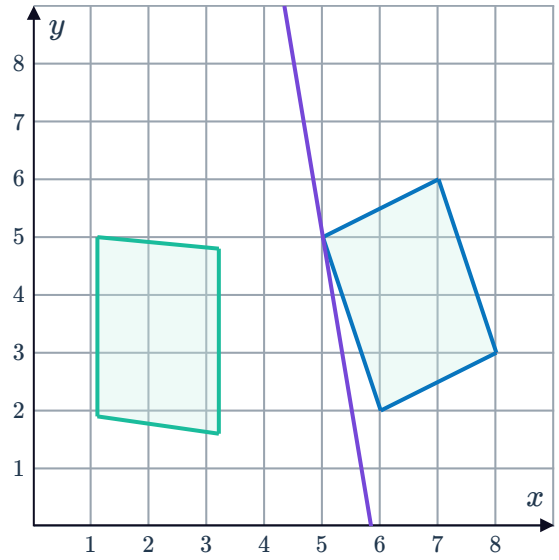
c



d



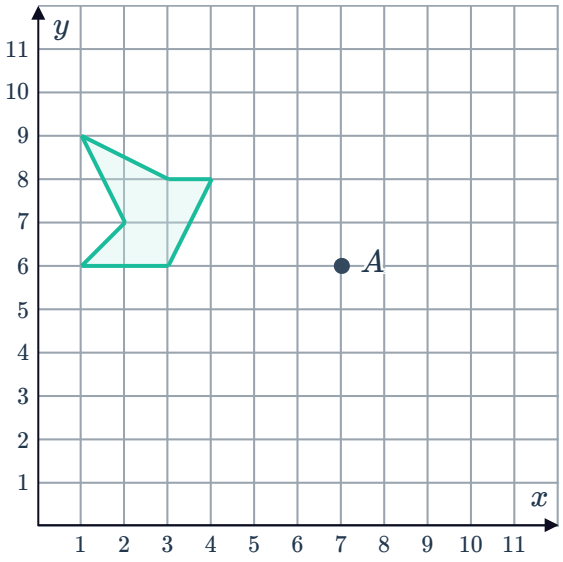
e



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4 Consider the following shape:

Draw the shape after it has been rotated by 180° clockwise about point A.

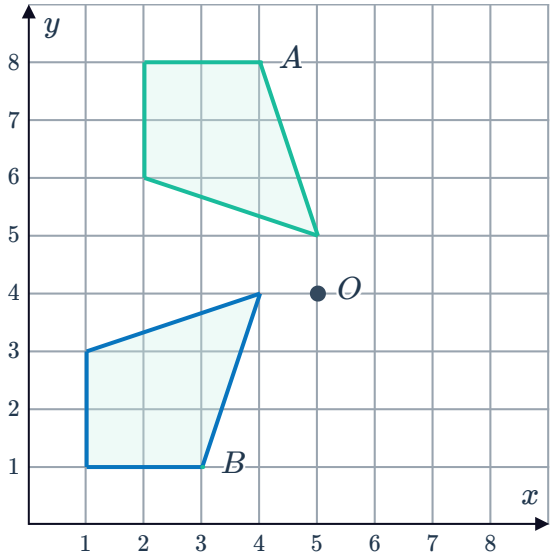


5 For each of the following:

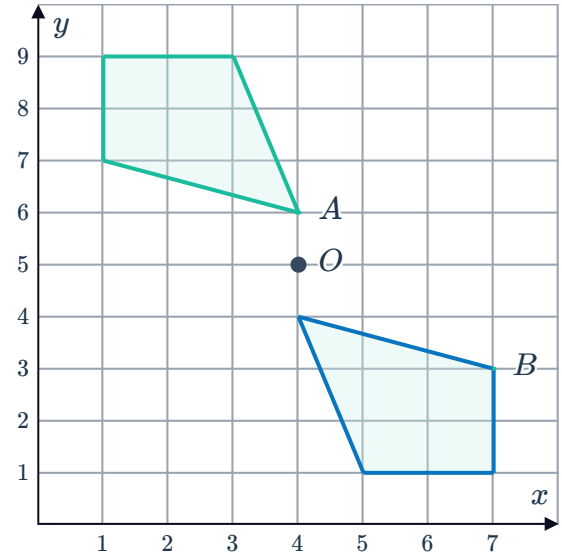
i Find the clockwise angle of rotation of shape A to shape B about point O.

ii Find the anticlockwise angle of rotation of shape A to shape B about point O.

a



b

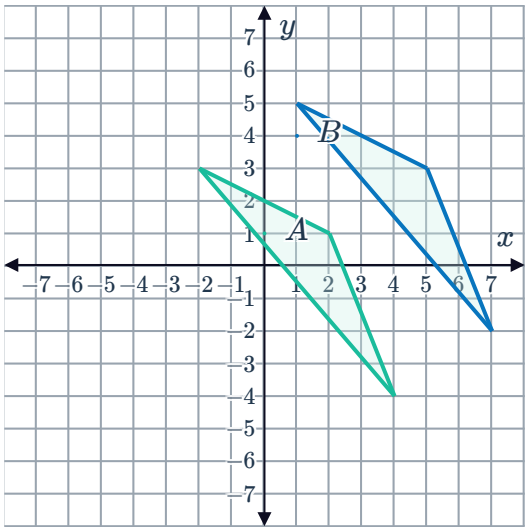


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6 Consider the figures shown:

a What type of transformation can take triangle A to triangle B?

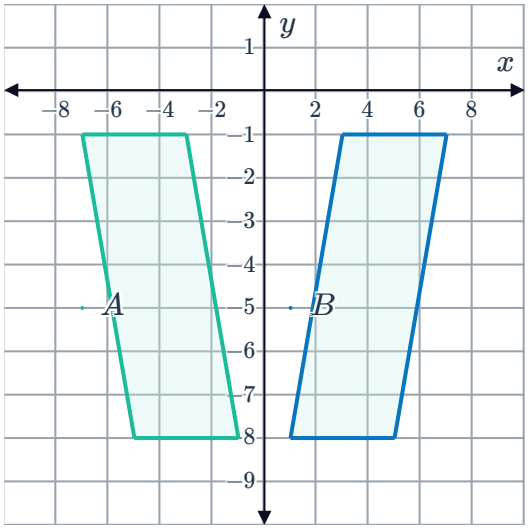
b Identify the transformation from triangle A to triangle B.



7 Consider the figures shown:

a What type of transformation can transform quadrilateral A to quadrilateral B?

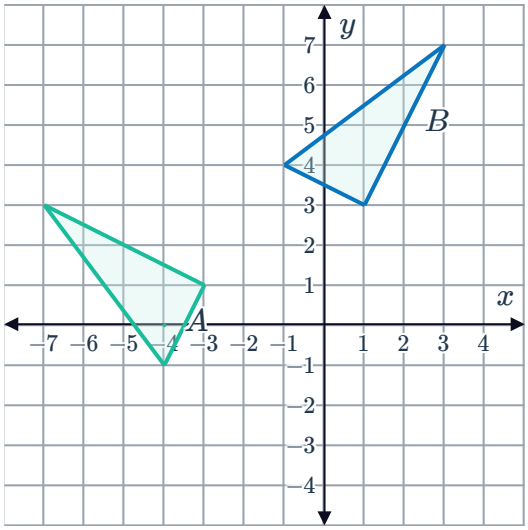
b Identify the transformation from quadrilateral A to quadrilateral B.



8 Consider the figures shown:

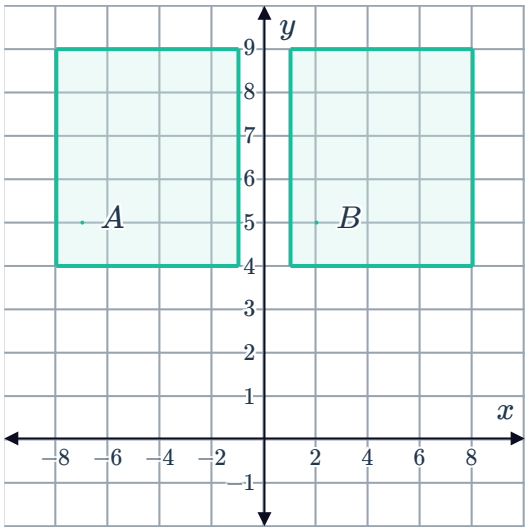
a What type of transformation can transform triangle A to triangle B?

b Identify the transformation from triangle A to triangle B.



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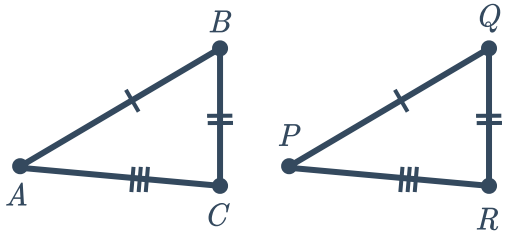
9 Consider the figures shown.
What are the possible types of transformations that could get rectangle A to rectangle B?



Congruence

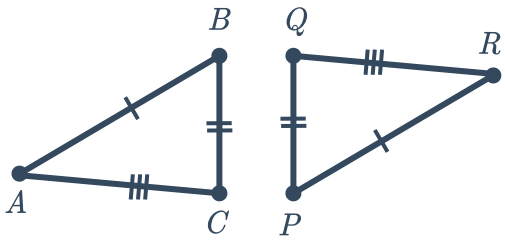
10 The given diagram shows 2 triangles that are translations of one another:

State the angle that has the same size as $\angle CBA$.



11 The given diagram shows 2 triangles that are rotations of one another:

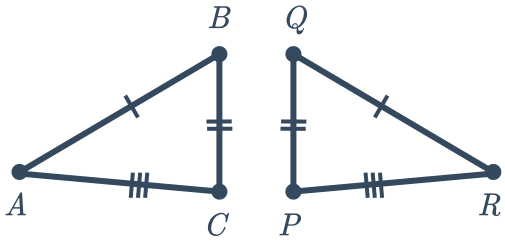
State the angle that has the same size as $\angle PRQ$.



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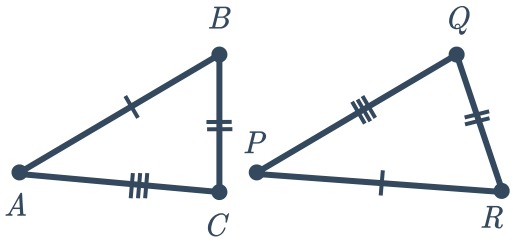
12 The given diagram shows 2 triangles that are reflections of one another:

State the angle that has the same size as $\angle RPQ$.



13 The given diagram shows 2 triangles that are translations and reflections of one another:

State the angle that has the same size as $\angle CBA$.



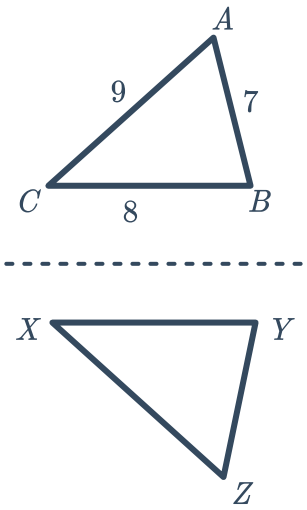
14 $\triangle ABC$ is reflected along the dotted line and its image XYZ is produced.

a Find the length of each side:

i XY ii YZ

iii XZ

b Describe the relationship between $\triangle XYZ$ and $\triangle ABC$.



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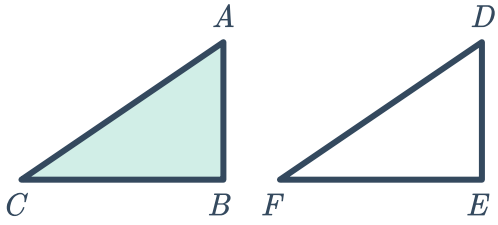
15 $\triangle ABC$ is translated to the right to form $\triangle DEF$:

a State which side of $\triangle DEF$ corresponds to BC.

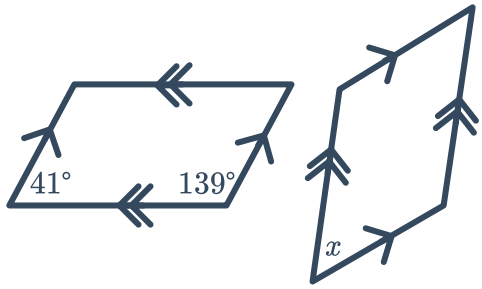
b State which angle of $\triangle DEF$ corresponds to $\angle BCA$.

c If $AC = 14$ cm, find the length of DF.

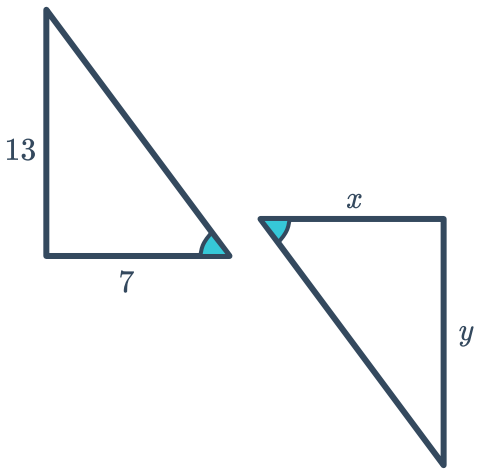
d If the smallest angle in $\triangle DEF$ is $\angle EFD$, state the smallest angle in $\triangle ABC$.



16 Given that the following shapes are congruent, find the value of x.

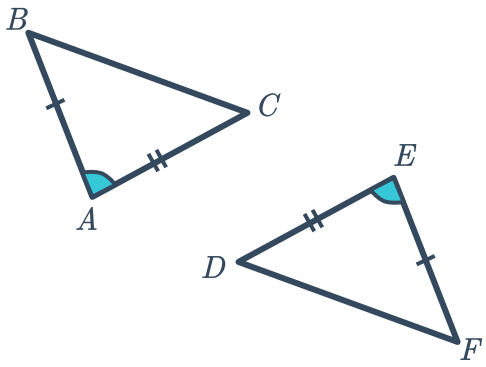


17 Given that the following triangles are congruent, find the values of x and y.

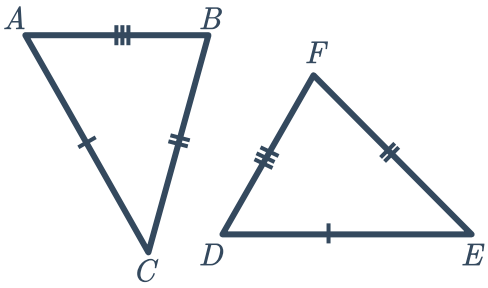


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18 Given that $\triangle BAC \equiv \triangle FED$, which angle of $\triangle FED$ corresponds to $\angle ABC$?



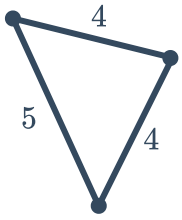
19 Which side is equal to AB in $\triangle DFE$?



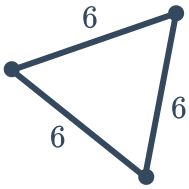
Bisectors

20 State whether the following triangles have a line of symmetry:

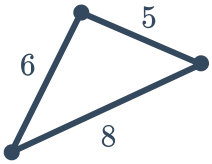
a



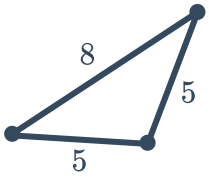
b



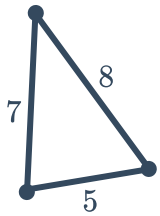
c



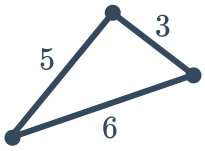
d



e



f



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21 Consider $\triangle PQR$ with midpoint X at side PR:

a State what type of triangle is $\triangle PQR$.

b State whether the following is correct.

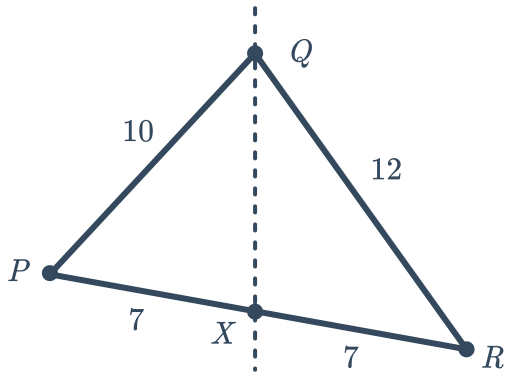
i $\angle QXR > \angle QXP$

ii $\angle QXR = \angle QXP$

iii $\angle QXR < \angle QXP$

c Is the line through Q and X a line of symmetry for $\triangle PQR$?

d Are $\angle RQX$ and $\angle PQX$ equal or not equal?



22 Consider $\triangle PQR$ with angle bisector PX of $\angle QPR$ and $QR = 12$:

a State what type of triangle is $\triangle PQR$.

b State whether the following is correct.

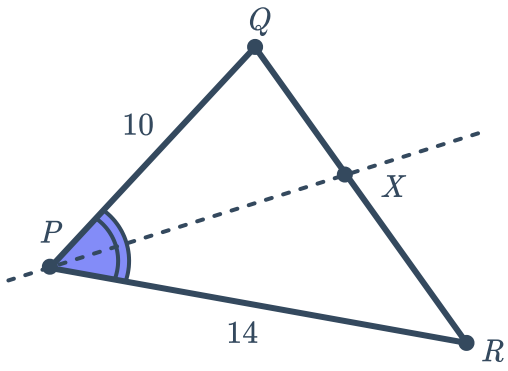
i $QX > RX$

ii $QX = RX$

iii $QX < RX$

c Is the line through P and X a line of symmetry for $\triangle PQR$?

d Are $\angle PXR$ and $\angle PXQ$ equal or not equal?



23 Consider the $\triangle PQR$ with midpoint X at side PR:

a State what type of triangle is $\triangle PQR$.

b Determine whether the following is correct.

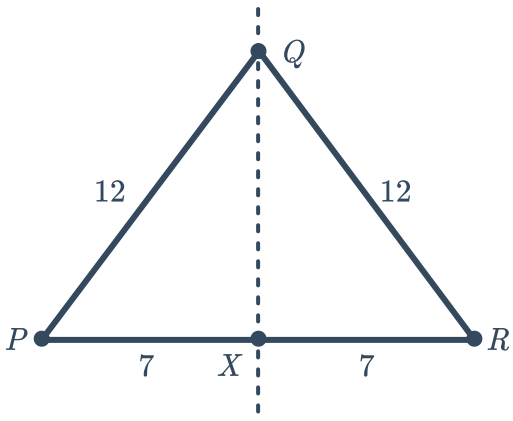
i $\angle QXR > \angle QXP$

ii $\angle QXR = \angle QXP$

iii $\angle QXR < \angle QXP$

c Is the line through Q and X a line of symmetry for $\triangle PQR$?

d Are $\angle RQX$ and $\angle PQX$ equal or not equal?



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24 Consider $\triangle PQR$ with angle bisector QX at $\angle PQR$:

a State what type of triangle is $\triangle PQR$.

b Determine whether the following is correct.

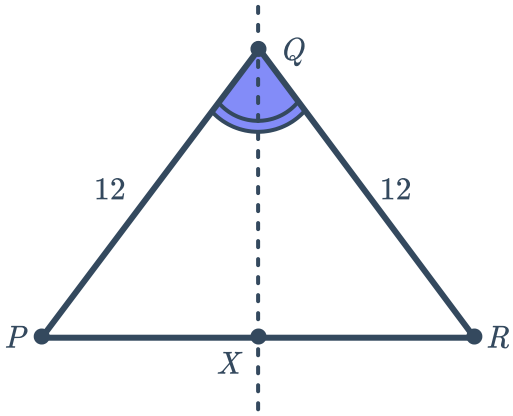
i $PX > RX$

ii $PX = RX$

iii $PX < RX$

c Is the line through Q and X a line of symmetry for $\triangle PQR$?

d Are $\angle QXR$ and $\angle QXP$ equal or not equal?



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